

Shaping Habits in Organizations: A Field Experiment*

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Abstract

We investigate whether organizations can shape work habits through monetary incentives. In a field experiment with 829 service technicians in 15 firms, we randomly allocated half of the technicians in each firm to a treatment group receiving bonuses for regularly performing sales activities for 12 weeks. We find a significant increase in sales activities not only during but also after the incentive phase. Using data from a post-experimental survey, we compare different behavioral channels. We find no evidence for increased automaticity, human capital acquisition, or signals about task priorities, but strong evidence for the role of acquired taste: Technicians in the treatment group report higher levels of intrinsic motivation to perform sales activities even after the incentive has been discontinued.

Keywords: HABIT FORMATION, TASTE ACQUISITION, TEMPORARY INCENTIVES
JEL classification: M12, M52, C93

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1 Introduction

Due to ongoing technological advancements and evolving market conditions, job profiles in organizations are subject to substantial change (Autor et al. 2024; Gibbs and Bazylik 2022; Berger et al. 2024; Böke et al. 2024). To encourage employees to adapt and assume new tasks, firms continuously revise performance targets, incentive structures, and training processes (Grabner and Moers 2013; Manthei, Sliwka and Vogelsang 2021; Buell, Cai and Sandino 2022; Hyun et al. 2022; Grabe and Sliwka 2025). However, for behavioral changes to have a lasting impact, they must persist even as the organization continues to evolve. A potential pathway to ensure this persistence is by fostering habits that integrate new behaviors into employees' daily routines. In fact, psychological research suggests that nearly half of an individual's daily actions can be considered habitual—that is, performed rather automatically without conscious deliberation (Wood, Quinn and Kashy 2002; Wood and Neal 2009), and it is therefore not surprising that many jobs involve a substantial share of routine tasks (Autor, Levy and Murnane 2003; Mihaylov and Tijdens 2019).¹ Despite the central role of routines in organizational functioning, relatively little is known about how to influence the formation of persistent habits in the workplace. Prior research on habit formation has largely examined the effectiveness of behavioral interventions (e.g. incentives, nudges, or reminders) on habits in non-work contexts, such as physical fitness and personal finance – domains that are characterized by different motivational structures compared to organizational settings (e.g. Buyalskaya et al. 2023; Charness and Gneezy 2009; Yang and Long Lim 2018; Allcott and Rogers 2014).

In workplace settings, however, the effectiveness of temporary incentives in fostering long-term habits remains unclear. As summarized in the review by Gneezy, Meier and Rey-Biel (2011), temporary incentives can help establish lasting habits in areas such as health, where individuals may come to appreciate the intrinsic benefits, such as improved well-being after initially engaging in the behavior due to external rewards. In work contexts, however, such intrinsic benefits may be less apparent to employees. Moreover, contract theory would predict that temporary incentives raise performance only temporarily, i.e. only as long as they lead to higher rewards but not afterwards. Beyond that, there is a risk that temporary incentives could lead to crowding-out effects, ultimately reducing employees' intrinsic motivation for the respective task below its original level (Deci 1971; Lepper, Greene and Nisbett 1973; Deci, Koestner and Ryan 1999; Gneezy and Rustichini 2000a,b). This effect may occur for example if the incentive

¹Note that such routine tasks "require methodical repetition of an unwavering procedure" (Autor, Levy and Murnane, 2003, p. 1283) and it has been argued that this very repetition is a key ingredient in habit formation as it allows behaviors to become automatic over time (Wood and Neal, 2009).

inadvertently signals that the task is undesirable (e.g., ‘the task is very difficult’) or if it conveys unfavorable social norms (e.g., ‘nobody else does it’) (Bénabou and Tirole 2003; Meier 2007; Sliwka 2007; Gneezy, Meier and Rey-Biel 2011; Alfitian, Sliwka and Vogelsang 2024).

In this paper, we investigate whether and how firms can foster the formation of habits at work. We follow a behavioral definition of habit formation, whereby habits are regularly repeated behaviors that are *insensitive to changes in the reward structure* (Dickinson 1985; Gillan et al. 2015). Therefore, if employees start engaging in a new behavior due to an extrinsic reward, but continue to do so after the incentive is removed, this indicates that a habit has formed.² We study this question in a field experiment with 829 employees in 15 technical service firms. These firms provide on-site installation and maintenance services on behalf of a large multinational company – hereinafter referred to as *MultiCo*. To understand how firms can support habit formation in the workplace, we randomized the implementation of a temporary monetary incentive that rewards sales-oriented behavior performed on a regular basis. Technicians in the treatment group received a €100 bonus for generating at least one sales lead per week over a period of four consecutive weeks. Sales leads are potential or existing customers who have given their consent to be contacted regarding the purchase of a specific product. The incentive was offered for a fixed period of 12 weeks, which was clearly communicated upfront to employees in the treatment group, allowing them to earn up to €300 in total. To support sales efforts, MultiCo seeks to encourage its technicians to leverage their on-site presence as an opportunity to advise clients on new or additional products and services. However, although the company has previously attempted to increase sales lead generation, these efforts have been largely unsuccessful, potentially due to the highly technical nature of the technicians’ vocational training and primary job roles.

We investigate two key pre-registered hypotheses: (i) the temporary incentive increases sales activities during its implementation, and (ii) this effect persists after the incentive has been discontinued. We test different cognitive mechanisms for the second hypothesis to better understand the formation of work habits and to provide companies with guidance for promoting them. For one, a behavioral shift may be due to increased *automaticity*. That is, while prior to the intervention many employees were not used to regularly talk to customers about potential sales, they now acquire the habit to do so. As argued in a rich literature in psychology (see e.g. Wood, Quinn and Kashy 2002; Wood and R  nger 2016; Verplanken and Orbell 2022), the fact that an activity is repeated frequently can lead to a certain automaticity of the behavior, i.e. an action is performed unconsciously

²Other definitions include additional assumptions on the cognitive foundations of habits, such as cue-response mechanisms (Mazar and Wood 2018; Wood and R  nger 2016) or (un-)conscious behaviors (Verplanken and Orbell 2022). See Volpp and Loewenstein (2020) and Chapman and Gneezy (2024) for a discussion regarding different definitions of habit formation.

and "out of habit" rather than through deliberate choice. As lined out by [Chapman and Gneezy \(2024\)](#), temporary incentives can also lead to a permanent shift in behavior due to *taste acquisition*: By repeatedly engaging in a new task, people may develop "a taste" for the new behavior, as they overcome initial aversion to certain aspects of the task. In our context, as the temporary incentive motivates technicians to perform sales activities more often, they may learn to enjoy customer interactions and continue to do so after the incentive has been discontinued. Another related channel is *human capital formation*: By repeatedly performing a task, employees learn from their experience, which reduces the costs of effort in the future ([Becker and Murphy 1988](#); [Jovanovic and Nyarko 1996](#)). In comparison to the prior explanation, the employees' motivation for the task itself remains unchanged, but the task becomes easier to perform due to larger task-specific human capital. Finally, the temporary incentive can also provide *direction*, as it may credibly reveal that their employer considers sales activities to be a promotable task or an important aspect of the overall strategy of the firm. This, in turn, may encourage employees to engage more persistently in these activities, for instance due to career concerns ([Holmström 1999, 2017](#)).

In line with our pre-registered hypotheses, we find a significant increase in sales leads not only during but also after the incentive phase. During the incentive period, the effect amounts to 31 percent of the pre-treatment mean. Although the effect after the incentive period is smaller in magnitude, it remains sizable at 15.5 percent higher performance relative to the the pre-treatment outcome. Successful sales leads – that is, leads resulting in actual sales – also increase by 16.8 percent during the incentive phase and and this effect persists during the post-incentive phase at 13.3 percent more successful leads. Using a post-experimental survey we compare the potential mechanisms of habit formation laid out in the above. We find consistent evidence for the role of acquired taste: Technicians in the treatment group report a significantly higher level of intrinsic motivation for sales activities after the incentive has been discontinued compared to those in the control group. Moreover, we observe a similar pattern for another customer-oriented task. It thus appears that the temporary incentive indeed had persistent performance effects because it led technicians to "learn to like" customer interactions and, in turn, achieve more sales. Further analyses show that the increase in sales activities did not reduce customer satisfaction. Despite the large upfront bonus payments for generating sales leads during the incentive phase, a back-of-the-envelope calculation suggests that the intervention led to a profit increase already within a few months after the end of the incentive phase.

Our contribution to the literature is twofold. First, we complement the rich literature on incentives in organizations.³ By showing that even temporary monetary incentives can be used to permanently align individual preferences and company goals, we underline the power of habits for organizational functioning. Previous studies on interventions targeting habit formation have focused on non-work behaviors such as exercise, commuting and health-related behaviors (Charness and Gneezy 2009; Larcom, Rauch and Willems 2017; Yang and Long Lim 2018; Allcott and Rogers 2014; Royer, Stehr and Sydnor 2015; Acland and Levy 2015; Gertler et al. 2018; Celhay et al. 2019; Gallani 2023). Second, our study also contributes to the debate on whether and where monetary incentives can undermine intrinsic motivation Frey and Oberholzer-Gee 1997; Gneezy and Rustichini 2000a,b; Kunz and Pfaff 2002; Bénabou and Tirole 2003; Sliwka 2007). To the best of our knowledge, our study is the first to provide (field) experimental evidence showing that monetary incentives can foster taste acquisition at the workplace (Chapman and Gneezy 2024; Loewenstein and Angner 2003) and thus *increase* rather than decrease intrinsic motivation after the incentive has been removed.

The remainder is structured as follows. Section 2 details the theoretical background. Section 3 and 4 describe the setting and the experimental design. Section 5 provides results on the temporary incentive effect and the underlying mechanism. Section 6 presents additional results on customer satisfaction and profit. Section 7 concludes.

2 Theoretical Background

As firms adjust their strategies in response to evolving technologies or market conditions, employees need to adapt to changing processes and acquire new skills. Such transformations present managers with new challenges, as they seek to guide and motivate their employees. Firms therefore need to provide guidance regarding expected behaviors and, in particular, the motivation to embrace new tasks and processes (Merchant and Van der Stede 2017). One way to address these challenges is through monetary incentives. It has often been argued that monetary incentives can effectively be used to align employee actions with company objectives. For instance, incentives can motivate employees by rewarding task completion (Ross 1973; Holmström 1979; Eisenhardt 1989) or provide direction by highlighting priority tasks (Manthei, Sliwka and Vogelsang 2023). Numerous empirical studies have indeed shown that monetary

³For surveys see, e.g., Prendergast (1999), Bandiera, Barankay and Rasul (2011), Lazear (2018), Mahlendorf and Vogelsang (2024).

incentives can raise performance during the period they are made available (e.g. Lazear 2000; Casas-Arce and Martinez-Jerez 2009; Friebe et al. 2017; Eyring and Narayanan 2018). In line with these findings, we expect that the temporary incentive leads to an upward shift in performance in the incentivized task during the time it is in place.

H1: The monetary incentive increases performance in the incentivized task during the incentive period.

Although the short-run effectiveness of monetary incentives is rarely disputed, it is not clear whether the long-term effect of temporary incentives is positive, neutral, or even negative. A key concern is crowding-out effects (Deci 1971; Lepper, Greene and Nisbett 1973; Deci, Koestner and Ryan 1999; Gneezy and Rustichini 2000a,b), where removing incentives lowers engagement below pre-incentive levels, for instance, due to reduced intrinsic motivation or detrimental effects on norms of behavior.⁴ In a workplace context, Alfitian, Sliwka and Vogelsang (2024), for instance, have recently shown that an incentive for perfect attendance backfired because it shifted social norms, making absenteeism more acceptable.⁵ However, for a crowding-out effect to occur, there must be a high level of intrinsic motivation or a strong social norm associated with the task at the outset. For novel or newly prioritized tasks this appears unlikely. Instead, we argue that for such tasks providing temporary incentives may help employees to align their attitude towards the behavior desired by the employer, as we will explain in the following.

Generally speaking, if a behavior persists after the initial incentive has been removed, this is often broadly defined as *habit formation* (Dickinson 1985). The underlying processes behind this behavioral pattern can be very diverse. To organize potential mechanisms for habit formation (see Volpp and Loewenstein (2020) or Chapman and Gneezy (2024) for recent reviews), we classify them according to the amount of cognitive deliberation

⁴In their review on the long-term effects of monetary incentives, Gneezy, Meier and Rey-Biel (2011) discuss two pathways towards a crowding-out effect: First, incentives can change the perceived locus of causality, i.e., an employee's intrinsic motivation to engage in a task is replaced by the external incentive (e.g. De Charms 1968; Huffman and Bognanno 2018; Brink, Lowe and Victoravich 2013). The second class of mechanisms encompasses signaling effects, where monetary incentives signal to the employee that a task is more difficult or less desirable than previously thought (Bénabou and Tirole 2003; Sliwka 2007; Cardinaels and Yin 2015; Danilov and Sliwka 2017).

⁵In a similar vein, a large literature on the incentives for whistle-blowing shows that rewards alter the perception of reporting financial misconduct from a (moral) obligation to act for one's own personal gain, which in turn reduces whistle-blowing in some situations (Brink, Lowe and Victoravich 2013; Berger, Perreault and Wainberg 2017; Andon et al. 2018).

that is involved in the action. On the end points of this continuum, we consider actions that are either *automatic* – which describes actions that are effortless, intuitive or even subconscious – or *deliberate* – which involves strategic and effortful thinking (see Figure 1).⁶

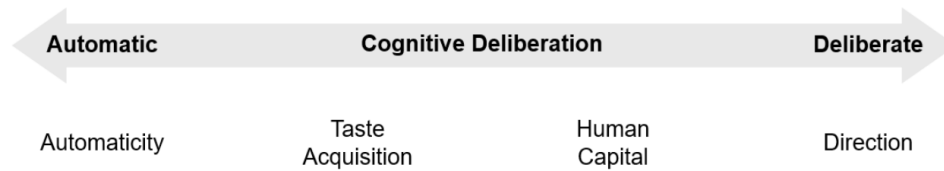


Figure 1: Potential Mechanisms for Habit Formation

Automaticity: A key component of many psychological definitions of habit formation is *automaticity* (Wood, Quinn and Kashy 2002; Verplanken and Orbell 2003; Wood and R  nger 2016). According to Mazar and Wood (2018), automatic habits are "goal-independent, unconscious, efficient, fast, and stimulus-driven" (p. 92). Behaviors that are often cited in these contexts are exercise and health-related (e.g. Charness and Gneezy 2009; Royer, Stehr and Sydnor 2015; Buyalskaya et al. 2023; Gallani 2023). Even if there was an initial conscious rationale behind some behaviors, they are often later on part of a subconscious routine that requires little to no reflection to be executed. Applied to the setting we study, consider a technician who receives multiple work assignments per day that require to install or troubleshoot specific technical equipment at the end-customer's home. Given that the technicians have repeated these processes hundreds of times, they have a standard routine that they follow for every work assignment. When these technicians now receive an incentive to also ask customers whether they may be interested in other products and services, they will likely initially do so to get the reward. However, over time this behavior may become part of their work routine. In this case, the technician at one point may no longer consciously engage with the customer in the hopes of generating a sales lead, but merely because it is part of their unconscious routine.

⁶Our classification maps well with the causes for management control problems described in Merchant and Van der Stede (2017). An automatic response (automaticity) corresponds to avoiding the control problem through automation. The remaining mechanisms correspond to each of the causes of the control problem, i.e. motivational problems (taste acquisition), personal limitations (human capital) and a lack of direction (direction).

Taste Acquisition: Following [Chapman and Gneezy \(2024\)](#), habits may also be formed as a result of *taste acquisition*. That is, continuous repetition of a behavior may lead a person to change their preferences towards the said behavior in a positive way. As a result, people consider the behavior to be more intrinsically rewarding.⁷ In contrast to automaticity, taste acquisition is a more deliberate process, because it often requires overcoming a certain level of personal resistance and once a preference shift has been realized the activity is pursued deliberately. Returning to our example, when a high monetary incentive leads the technicians to engage with customers more frequently, it may enable them to discover the positive side of customer interaction ([Melchionne 2007](#); [Loewenstein and Angner 2003](#)).

Human Capital: While taste acquisition is focused on people changing their motivational stance towards a particular activity, it is further possible that the incentive leads to an increase in human capital either through stronger incentives to acquire knowledge or through learning-by-doing. As employees refine their skills through continuous practice, it becomes easier for them to engage in the behavior in the future ([Arrow 1962](#); [Becker 1965](#); [Jovanovic and Nyarko 1996](#); [Wyatt and Frick 2010](#); [Manthei, Sliwka and Vogelsang 2021](#)). This explanation for habit formation is also reflected in classical models of consumer habits in economics ([Becker and Murphy 1988](#); [Stigler and Becker 1977](#)). According to these models, past frequency of the behavior increases the stock of “consumption capital” ([Becker and Murphy 1988](#), p. 667). In this case, the long-term effect of the incentive is driven by an increase in knowledge and skills, while the underlying motivation remains unchanged. For example, when technicians receive monetary incentives for generating sales leads, they may be more motivated to engage with learning materials that enhance their approach. Similarly, as they gain experience in customer service, their conversational skills improve, leading to higher customer satisfaction ratings and – in turn – more sales. However crucially, if sales activities are no longer lucrative or become increasingly difficult at a certain point, employees will scale back their efforts accordingly.

Direction: Finally, it is possible that the temporary incentive system sends a signal about the firms priorities or overall strategy. In the spirit of [Holmström \(2017\)](#), “Putting money behind a measure conveys a stronger message of what is expected” (p. 1772). In that sense, technicians may infer from the bonus for sales activities that engaging

⁷One might argue that acquiring tastes is conceptually similar to the reduction of cognitive dissonance that occurs whenever someone engages in an undesirable activity for the sake of a desirable outcome. ([Festinger 1957](#); [Harmon-Jones and Mills 2019](#)). According to the theory of cognitive dissonance, employees might rationalize their sales activities by re-framing them as more useful for the customer or by selectively remembering pleasant customer interactions. However, the theory further predicts that dissonance reduction is substantially reduced when the rewards for the unpleasant activity are sufficiently large ([Festinger and Carlsmith 1959](#)). Additionally, the theory provides no explanation for long-term effects once the incentive has been removed.

in sales behavior is of high importance for the firm. In turn, such behavior can then appear more relevant for career progression and wage increases, in line with the classical model of career concerns (Holmström 1999).⁸ In summary, we argue that temporary incentives can lead to habit formation in the workplace and we test each of the presented mechanisms and alternative explanations for a persistent behavioral shift using data from a post-experimental survey.

H2: The monetary incentive increases performance in the incentivized task after the incentive has been discontinued.

3 Institutional Setting

Our sample consists of all active technicians from 15 medium-sized technical service firms who install and maintain products and services on behalf of MultiCo. MultiCo both employs its own technicians and works with these service firms as sub-contractors. MultiCo's field of work is divided into 37 regions. Individual employees at the subcontractors are typically assigned to specific local areas within a region and receive their work schedule every morning from their local dispatcher. While a few assignments can be performed remotely, most of the work is done at the client's site. Even though MultiCo's employees operate in the same infrastructure, interaction between employees, even within the same team members is very limited.

To leverage technicians' customer contact, MultiCo has increasingly instructed technicians to use customer visits as an opportunity to promote and sell additional products and services. While a few products can be provided directly from the technician's mobile inventory, the larger volume is processed by MultiCo's customer service. If a customer is interested in additional products or services, employees create a *sales lead* that is handled by MultiCo's customer service. If the customer service agent manages to sell an additional product to the customer, the lead is considered *successful* and the technician who created the lead receives a commission for the successful sale. Employees have access to a list that shows which products are incentivized and how large the commission for a specific product is. Products with a higher profit margin also lead to higher commissions.⁹ As the processing of generated sales leads and actual sales

⁸Similarly, the incentive may cause what the accounting literature terms strategy surrogation, setting a strong focus on the specific performance measure used (Choi, Hecht and Tayler 2012, 2013; Wang, Cheng and Chang 2023).

⁹Commissions can range from less than €10 to up to €175 per product sold. A single sales lead can result in the sale of more than one product. On average, 38.71% of sales leads during the pre-period are successful. The average commission earned from each successful sales lead for an incentivized product is €22 in that time frame.

do not occur immediately after the customer visit, there may be substantial time lags until a bonus payout occurs. Commissions for successful leads prior to the experiment are typically paid out within eight weeks, with 75% issued during that time frame. However, certain products and services have longer wait times due to their complexity or the need for extensive paperwork.

To support the service employees in identifying sales opportunities and approaching the customers, MultiCo offered online sales trainings to all employees before and at the time of the experiment. Training sessions took place outside of regular working hours.¹⁰ However, neither the existing bonus scheme for successful sales nor the online trainings led to a substantial increase in technician's sales activities.

4 Experimental Design and Data

We randomize the implementation of a temporary bonus scheme to foster regular sales activities among technicians. Employees in the treatment group receive a bonus of €100 for generating at least one sales lead per week for four consecutive weeks. Importantly, different from the existing sales commission scheme (which remains in place), the sales lead in this new temporary bonus scheme does not have to be successful for employees to receive the bonus. The intervention runs for twelve weeks, such that employees can receive the bonus up to three times, resulting in a bonus of up to €300.¹¹ After the end of the twelve week intervention period, we observe individuals for a twelve week post-intervention period in order to assess if there is a persistent behavioral change regarding the reporting of sales leads.¹²

The experiment started in March 2023. Employees were informed up to 1.5 weeks prior to the experiment.¹³ We randomly assigned 923 individual employees (from the total of 15 firms) to treatment and control groups using stratified randomization based on the employee's company and their own previously reported sales leads (dummy equal to one if they would have received at least one bonus in the last 12 full weeks and zero otherwise, and a dummy equal to one if they reported sales leads since October 2022 and zero otherwise). Hence, we randomize treatment assignment *within* firms to

¹⁰As only few employees took up the offer, the training sessions were temporarily discontinued around one month after the start of the experiment.

¹¹A service technician receives a gross wage of around €2,900 per month. Thus, they can get around 3.4% extra through the incentive.

¹²For fairness reasons so that no employee is disadvantage based on the random assignment, technicians in the control group also receive the new bonus scheme for twelve weeks. This period started two weeks after the end of the post-intervention period.

¹³Please see Figures 4 to 6 in the Appendix for the wording of the email announcements for the respective groups.

be unaffected by potential differences in firm-specific time trends. To assure that we consider only employees who benefit from the bonus, we restrict our primary analyses to the 829 employees who remained in the firm until the end of the post-intervention period.¹⁴

As Table 1 shows, the sample is well balanced with respect to our main outcomes and across firms. Service technicians in the incentive group as well as the control group report around 1.3 sales leads per week on average in the pre-period 0.5 of which were successful on average. Also, the regularity with which sales leads are reported is not statistically significantly different between groups. In both groups, the average employee would have received about 0.5 bonuses in the pre-period if the scheme had been in place already at this point in time.

¹⁴As Table 6 in the Appendix shows, there is no evidence for selective attrition based on time or prior sales performance. The table shows that 9.8% of service technicians in the incentive group and 10.6% in den control group left their firm until the end of the post-intervention period. During the incentive phase, 4.6% (incentive group) and 4.8% (control group) left their employer. In the first four weeks after the incentive phase, 1.3% and 0.6% left (p -value=0.313).

Table 1: Balance Check

	<i>Incentive Group</i>		<i>Control Group</i>		p-value
	Mean	S.D.	Mean	S.D.	
Weekly Leads Pre	1.271	1.741	1.333	1.981	0.637
Weekly Succ. Leads Pre	0.503	0.751	0.504	0.729	0.981
Bonuses Pre	0.493	0.694	0.525	0.712	0.504
Company 1	0.053	0.224	0.048	0.215	0.770
Company 2	0.075	0.263	0.080	0.271	0.772
Company 3	0.022	0.146	0.027	0.161	0.639
Company 4	0.149	0.357	0.128	0.335	0.388
Company 5	0.053	0.224	0.053	0.225	0.980
Company 6	0.082	0.274	0.094	0.293	0.519
Company 7	0.137	0.344	0.140	0.348	0.887
Company 8	0.031	0.174	0.029	0.168	0.853
Company 9	0.058	0.233	0.063	0.243	0.750
Company 10	0.050	0.219	0.051	0.220	0.981
Company 11	0.058	0.233	0.051	0.220	0.664
Company 12	0.123	0.328	0.126	0.332	0.885
Company 13	0.038	0.193	0.036	0.187	0.871
Company 14	0.022	0.146	0.024	0.154	0.804
Company 15	0.050	0.219	0.048	0.215	0.891
Observations	416		413		829

Note: This table reports summary statistics separately for treatment and control group. Additionally, we report p -values for t-tests comparing the means of the continuous variables and test of proportions for the dummy variables, respectively. The table is based on out balanced sample, i.e. only including individuals who are still there at the end of our observation period.

5 Results

5.1 Effect on Sales Leads

To investigate the effect of our bonus intervention on the technicians' behavior, we first examine the number of reported sales leads over the span of our observation period. Figure 2 depicts the number of sales leads for the treatment and control group from the beginning of 2023 until the end of the post-intervention period.

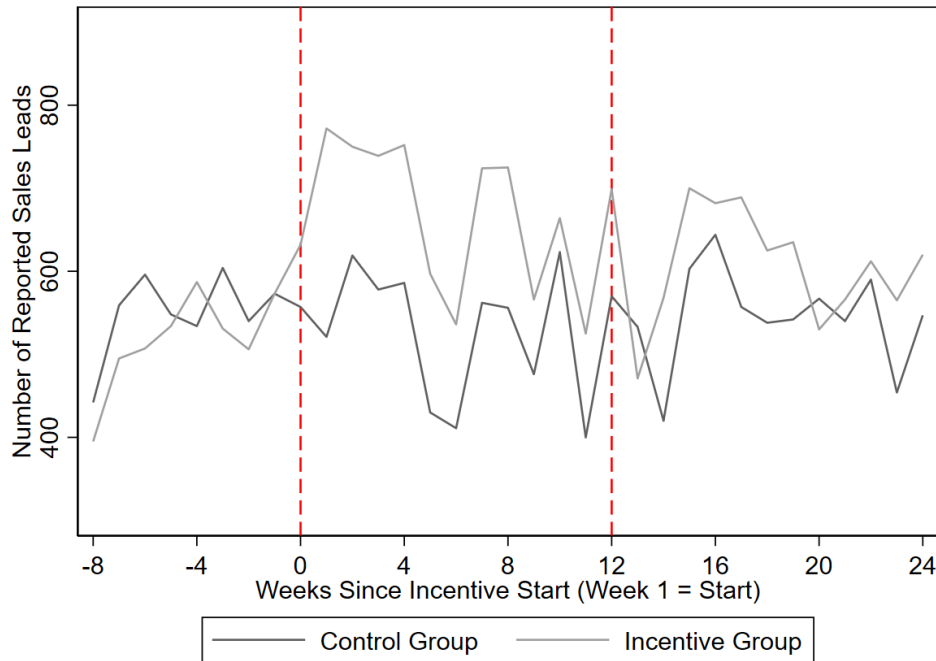


Figure 2: Sales Leads over Time

Note: This graph plots the number of sales leads by treatment group over the weeks since the incentive start (first week with incentive is week 1). The graph starts with the beginning of 2023. The red line at week 0 marks the last week before the start of the incentive phase. The employees were already informed about the upcoming bonus during week -1. The red line in week 12 marks the last week of the incentive phase. We use the balanced panel, i.e. only include individuals who are still there at the end of our observation period.

As Figure 2 shows, there is a rather sharp increase in reported sales leads in the treatment group when the treatment is implemented¹⁵ and the number of reported leads stays larger in the treatment than in the control group during the 12 week treatment phase. But the number of reported leads also tends to be higher on average even after the end of this period.

¹⁵Recall that the treatment was announced about 1.5 weeks before the bonus was active which started to be payoff relevant in week 1. There seems to be an anticipatory effect already in week 0 which may be the result of technicians already trying out how to approach customers about additional services before the actual start of the incentive period.

Table 2: Effect on Number of (Successful) Sales Leads

	<i>Sales Leads_{it}</i> (1)	<i>Successful Leads_{it}</i> (2)
$Treat_i \times Incentive_t$	0.396*** (0.073)	0.084** (0.035)
$Treat_i \times Post\ Incentive_t$	0.197** (0.100)	0.067* (0.038)
<i>p</i> -value Inc=Post	0.014	0.569
Time Fixed Effects	Yes	Yes
Individual Fixed Effects	Yes	Yes
Clustered at	Individual	Individual
Number of Clusters	829	829
Observations	27,357	27,357
Adjusted R-squared	0.565	0.409

Note: This table reports results of a difference-in-differences regression of the number of reported sales leads (column 1) or successful sales leads (column 2) on being part of the treatment group, i.e. incentive group, during as well as after the incentive phase in comparison to the pre-incentive phase. The data is on the individual-week level. The regression is based on the balanced panel, i.e. only include individuals who are still active for the firm at the end of our observation period. We include individual as well as week fixed effect. Standard errors are clustered at the individual level, and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Column (1) in Table 2 shows the results of difference-in-differences regression of the number of sales leads on a treatment dummy during the intervention and post-intervention periods.¹⁶ We see a strong effect of the incentive during the incentive period. The treatment increases the number of sales leads per employee by around 0.4 on average per week, which corresponds to a 31% increase relative to the pre-period average of 1.3 sales leads per week. The effect is smaller in magnitude but still sizeable at 15.5% in the post-intervention period.¹⁷

As the technicians received the bonus for reported sales leads independent of the success of these sales leads, it is conceivable that technicians simply increase the number of reported sales leads even when these customers do not have an intention to buy further products or services. However, as column (2) in Table 2 shows, the treatment also increased the number of successful sales leads per technician and week on average by 0.084, which is around 16.8% in comparison to the pre-period mean of 0.5 successful sales leads. Again the coefficient is somewhat smaller for the post-intervention period

¹⁶Table 7 in the Appendix shows the results of a cross-sectional comparison between the treatment and control groups during and after the incentive phase, confirming the main results.

¹⁷The difference between the effect in the incentive and post-incentive period is significant (Wald-test p -value=0.014).

but technicians reported sales leads still lead to 0.067 or about 13% more successful sales even after the monetary incentive has expired.¹⁸ Thus, while not each additional sales lead generated through the incentive is successful, the incentive also translates into a sizable increase in successful sales leads during and after the incentive phase.¹⁹

We further investigated whether the incentive for regularly reporting sales leads not only increased the number of (successful) sales leads but also their regularity. To test that, we also consider the number of bonuses a technician would have received (i.e. when all had been part of the bonus group all the time; that is the number of consecutive four week periods with a sales lead in each) as well as the number of weeks with at least one sales lead as outcome variables. Table 9 in the Appendix shows the results. While the number of bonuses a technician would have received only significantly increases during the incentive period, the number of weeks with at least one sales lead increases both during the incentive as well as the post-intervention period. Thus, even though technicians do not seem to try anymore to reach exactly four weeks in a row with sales leads, they still create sales leads more regularly. Overall, we find evidence for habit formation in response to temporary incentives that is persistent beyond the incentive period. Thus, our results support *H1* and *H2*.

5.2 Mechanisms

To understand the behavioral mechanisms underlying our findings, we conducted a survey during the post intervention phase. The survey contains a set of items for each of the potential behavioral channels described in Section 2, namely, *automaticity*, *taste acquisition*, *human capital formation*, or *direction*. All participants were assured at the beginning of the survey that individual answers were pseudonymized and would, under no circumstances, be shared with supervisors or MultiCo employees.

We used the automaticity subscale of the self-reported habit index (Verplanken and Orbell 2003) as a measure for *automaticity*, which is a standard scale with high predictive validity that has been used in a multitude of studies (Gardner et al. 2012; Gardner, de Bruijn and Lally 2011; Gardner, Rebar and Lally 2022).²⁰ To assess *taste acquisition*

¹⁸The percentage of successful sales leads in the treatment group during the post-period is 38.84%. The average commission earned per successful sales lead for an incentivized product in this timeframe is €22.51. Thus, there is only a very slight change compared to the pre-period, which had 38.71% successful sales leads and an average commission of €22.

¹⁹Table 8 in the Appendix shows the results when including all service technicians we initially randomly assigned. Their number of sales leads is reported as 0 for the weeks in which they were no longer able to report sales leads.

²⁰Items are: "Talking about additional services/products when visiting customers is something . . ." 1. ". . . I do automatically", 2. ". . . I do without having to consciously remember", 3. ". . . I do without thinking", 4. ". . . I start doing before I realize I'm doing it" (Scale: 1 (do not agree at all) - 7 (Fully agree); Cronbach's Alpha: $\alpha = 0.91$).

we use a common measure of *intrinsic motivation* for a task using three items from the interest/enjoyment sub-scale of the Intrinsic Motivation Inventory (Ryan, Mims and Koestner 1983) adapted to the sales task.²¹ In order to assess treatment effects on *human capital* acquisition, we conducted a short quiz to test the employees' knowledge about sales activities. All questions were based on existing training material of the firm that was regularly used to teach technicians how to effectively generate promising sales leads.²² Finally, we measured the perceived importance of sales activities to determine whether the temporary incentives provided technicians with *direction* on work priorities expected of them. These questions capture, for instance, whether technicians perceive sales activities to have a high priority for managers or the firm in general.²³

Table 3: Summary Statistics for Survey Responses

	<i>Incentive Group</i>		<i>Control Group</i>	
	Mean	S.D.	Mean	S.D.
Automaticity Index	4.749	1.839	4.541	1.768
Intrinsic Motivation	4.755	1.752	4.231	1.526
Knowledge	3.481	1.032	3.476	1.024
Perceived Importance	5.594	1.432	5.359	1.399

Note: This table reports summary statistics for the survey responses. Responses for items on automaticity, intrinsic motivation, and perceived importance are measured on Likert scales ranging from 1 (lowest) to 7 (highest). For each construct, we calculate the mean of all corresponding items. Knowledge is assessed as the number of correct answers in a short quiz consisting of five questions.

The survey was distributed to all service technicians from 12 out of 15 companies who were active at that time (4 weeks after the end of the incentive phase).²⁴ The invitation was adapted from previous MultiCo surveys and did not mention sales activities or incentives as a topic of interest. Note that the survey specifically asks about

²¹Items are: "Talking about additional services/products when visiting customers is something . . ." 1. ". . . that gives me great pleasure", 2. ". . . that I enjoy ", 3. (reversed) ". . . that I find boring" (Cronbach's Alpha: $\alpha = 0.70$)

²²We measure that using five closed-ended questions regarding how to generate sales leads, e.g. "To make sure the customer doesn't feel pressured into making a purchase, I can use phrases like "Should we take a look?" or "Should we try this?". As a further validation we also repeatedly posted the questions (in the original German wording) to ChatGPT-4o prompting it to consider best practices and scientific insights when answering. In all cases, the answers provided were identical to those deemed correct in the knowledge quiz. When asked to assign a certainty level to these answers, the confidence levels ranged from 85% to 100%.

²³The items are: "Talking about additional services/products when visiting customers is something . . ." 1. ". . . that my boss expects of me", 2. ". . . that is one of the important tasks of my job", 3. ". . . that is demanded of me" (Cronbach's Alpha: $\alpha = 0.67$)

²⁴Due to legal regulations, each firm had to grant the research team the authority to directly contact their employees. While 12 of them agreed, the remaining 3 firms did not allow us to send out the surveys. Given that the firms that declined their participation are among the smallest firms in the sample (we contacted 92% of all technicians in our sample) and as treatment assignment was randomized within firms, we do not consider this a threat to validity.

the technicians' experiences in the last two weeks, i.e., only in the post-treatment period. Participants were paid a participation fee (€8) conditional on completing the survey. The response rate was around 30%.²⁵ We do not find that the share of survey participation differs between treatment and control group (p -value = 0.754). Table 3 shows descriptive statistics for the four measures for the treatment and control group respectively.

Table 4 presents the regression results where the standardized survey constructs are regressed on the treatment dummy. We do not find significant effects of the treatment on perceived automaticity in behavior, the direction through the incentive that sales is an important task, or human capital acquisition. However, we find a significantly higher level of intrinsic motivation for sales activities in the incentive group. In line with recent ideas, such as those in [Chapman and Gneezy \(2024\)](#), the monetary incentives appear to have led employees to acquire a taste for the target behavior, which persisted even beyond the incentive period.

Table 4: Survey Results

	<i>Automaticity_i</i> (1)	<i>Taste Acquisition_i</i> (2)	<i>Human Capital_i</i> (3)	<i>Direction_i</i> (4)
<i>Treat_i</i>	0.115 (0.127)	0.315** (0.126)	0.005 (0.126)	0.166 (0.126)
Observations	248	245	253	252
R-squared	0.003	0.025	0.000	0.007

Note: This table presents the results of a regression analysis examining the effect of being in the treatment group (i.e., the incentive group) on measures of Automaticity, Taste Acquisition, Human Capital and Direction. Automaticity is measured using the four Items from the Automaticity sub-scale of the Self-Reported Habit Index in [Verplanken and Orbell \(2003\)](#). Taste acquisition is measured using three items from the Intrinsic Motivation Inventory by [Ryan, Mims and Koestner \(1983\)](#). Human Capital is measured using a quiz consisting of four questions based on MultiCo's training material. Direction is measured using three items assessing the perceived importance of the task. Questions are answered on a scale from 1 = "fully disagree" to 7 = "fully agree". An exception is our measure for human capital, where the answers are always "yes" or "no". All measures are calculated as the means of all items, which are standardized using z-scores. The data is at the individual level. There are slight differences in the number of observations across the constructs, as not all survey participants answered every item. See Table 10 in the Appendix for results based only on participants who answered every item. Robust standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

To test this *taste acquisition* channel further, we can make use of additional survey items about technicians' preferences for another customer-oriented task, namely calling customers before visiting them. Employees are strongly encouraged to contact the customer prior to the appointment to personally introduce themselves to the customer and to get a better understanding of their issue. We indeed observe a similar increase in

²⁵However, not all participants answered every item. See Table 10 in the Appendix for results based only on participants who answered every item.

intrinsic motivation for contacting customers before the visit.²⁶ Therefore, this findings suggest that temporary incentives even created spillover effects on other customer oriented tasks. This supports the view that many technicians initially viewed the sales task more as a nuisance. The monetary incentive then provided material incentives to perform the task and seek out customer contact at a regular frequency. In turn, this seems to have led a sufficient number of technicians to realize that customer contact can be enjoyable.

While the treatment coefficients in the regressions for the *automaticity* and *direction* measures are statistically insignificant, the point estimates are not sufficiently small to rule out the relevance of these mechanisms altogether. But concerning the *direction* channel, it is important to note that the respective survey responses show that employees are well aware at the outset that sales has a strong priority: As Table 3 shows, even employees in the control group rate the perceived importance of the sales task as rather high, with a mean of 5.4 on a scale from 1 (lowest) to 7 (highest). Thus, a lack of understanding of the task's importance appears unlikely to explain why employees do not engage more in sales activities in the first place.

6 Further Results

6.1 Customer Feedback

We conducted additional analyses on two further aspects: customer satisfaction and sales profit. Although we did not formulate specific hypotheses regarding these aspects, they are essential for evaluating the success of this bonus scheme. While our incentives increase the number of sales leads, they may also lead to dissatisfaction of customers in case they feel overly pressured into agreeing to talk about additional products and services. Hence, we analyze the effect of our incentive on customer satisfaction ratings of the technicians' visits. After the technicians complete their visit, customers receive an automated feedback request by MultiCo where they are asked to rate the technicians visit on a five-star rating scale ranging from 1 star (worst) to 5 stars (best).

Table 5 shows the results of a difference-in-differences regression of customer ratings on being in the incentive group during the incentive period or in the post-intervention period. We do not find a decrease in the average star rating for the visits. If anything, the coefficient is even positive albeit insignificant (column (1)). The number of visits that

²⁶See Table 11 in the Appendix.

Table 5: Effect on Customer Satisfaction Ratings

	<i>Star Rating_{it}</i> (1)	<i>Number Lowest Rating_{it}</i> (2)	<i>Number Top Rating_{it}</i> (3)
$Treat_i \times Incentive_t$	0.040 (0.029)	-0.021* (0.012)	-0.104 (0.073)
$Treat_i \times Post\ Incentive_t$	0.014 (0.028)	-0.009 (0.014)	-0.108 (0.083)
<i>p</i> -value Inc=Post	0.365	0.337	0.959
Controls	Yes	Yes	Yes
Time Fixed Effects	Yes	Yes	Yes
Individual Fixed Effects	Yes	Yes	Yes
Clustered at	Individual	Individual	Individual
Number of Clusters	783	783	783
Observations	19,693	19,693	19,693
Adjusted R-squared	0.290	0.148	0.310

Note: This table reports results of a difference-in-differences regression of the customer satisfaction rating (column (1)), number of lowest possible ratings (column (2)) or number of highest possible ratings (column (3)) on being part of the treatment group, i.e. incentive group, during as well as after the incentive phase in comparison to the pre-incentive phase. The data is on the individual-week level. The regression is based on the balanced panel, i.e. only include individuals who are still there at the end of our observation period. We control for the share of being on-time and fulfilling the task of the individual in the week. Furthermore, we include individual as well as week fixed effect. Standard errors are clustered at the individual level, and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

received the lowest rating is even marginally significantly reduced during the incentive phase (column (2)). However, also the coefficient for the number of five star ratings is negative but insignificant (column (3)). Thus, overall we do not find evidence for a change in customer satisfaction due to the incentive scheme.

6.2 Profit

A natural question to assess the success of the incentive scheme is whether the increase in the number of successful sales leads can offset the additional costs for the bonuses and lead to higher profits. To explore this, we estimate the monthly net profits generated by a technician as total value added of the products sold based on sales leads reported by this technicians in this month (also if the actual sale was later in time) subtracting the total commission paid for the sales leads in that month. Profit margins are based on a simplified report we received from MultiCo.

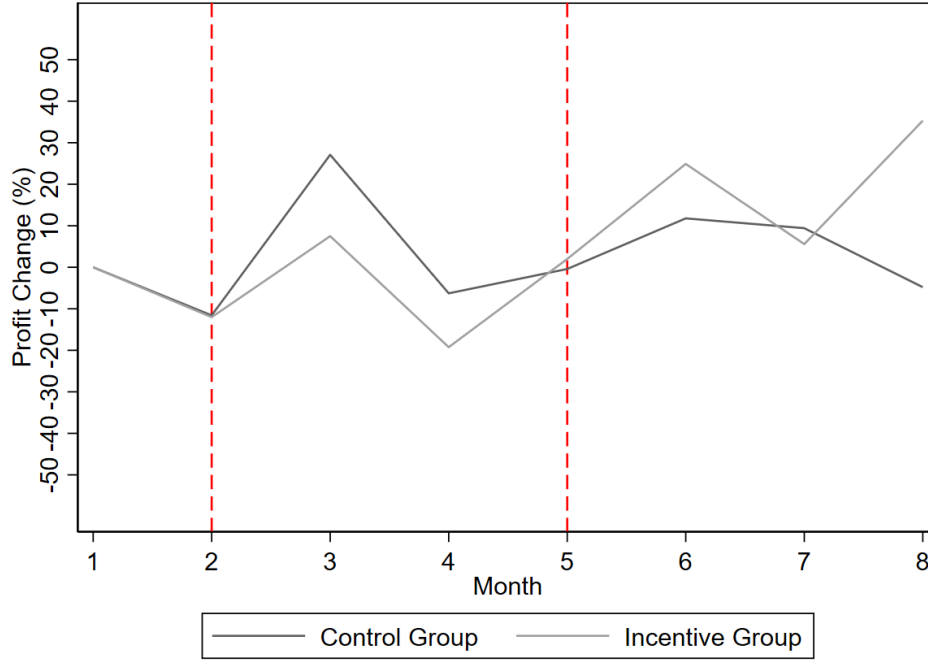


Figure 3: Sales Profit over Time

Note: This graph plots the sales profit change in percent by treatment group over the calendar months of 2023. The profit is normalized such that the profit is the same for both groups in January (month 1). The red line at month 2 marks the last month before the start of the incentive phase. The red line in month 5 marks the last month of the incentive phase. We use the balanced panel, i.e. only include individuals who are still there at the end of our observation period.

Figure 3 shows the percentage change in profits relative to the January profits separately for incentive and control group over time. As the Figure shows, the incentive group generates lower profits during the incentive phase as the costs of the incentive scheme outweigh the benefits during this phase. In the post-incentive period, however, the incentive group tends to generate larger profits as technicians in the treatment group continue to provide more successful sales leads as already shown above, but the firm no longer has to pay the additional bonus payments. The corresponding regression results are shown in Table 12 in the Appendix. By a simple back-of-the-envelope calculation the break-even point is attained in month 8. As the control group received bonuses starting at the beginning of September (month 9), we cannot observe how long the positive effect on profit lasts. However, as it appears very unlikely that crowding-out effects occur with a strong time lag the temporary bonus is very likely to be profitable overall.

7 Conclusion

We present evidence showing that workplace habits can be shaped through temporary monetary incentives. In our setting, employees continued to perform a task at a higher intensity even after the temporary incentive was removed. Thus, our findings complement prior research on the long-term effects of temporary incentives and management controls in general. While the previous literature has stressed the danger that monetary rewards may crowd out intrinsic motivation, we show that incentives can also lead to a "crowding-in" of task motivation at the workplace. We furthermore provide evidence supporting the role of taste acquisition ([Chapman and Gneezy 2024](#)) in this process: when monetary incentives motivate employees to perform a task more frequently, they may learn to like it and then perform it without further external reinforcement.

It is important to note that these results do not contradict previous findings on potential detrimental effects of temporary incentives. The results rather suggest that one should consider the prior task motivation or existing social norms. As laid out in the above, previous studies documenting detrimental effects of incentives typically investigated settings in which intrinsic motivation was rather high at the outset ([Deci 1971](#); [Gneezy and Rustichini 2000a,b](#)) or there were strong prior favorable social norms ([Alfitian, Sliwka and Vogelsang 2024](#)). When this is the case, detrimental effects indeed appear more likely to occur as there is more scope to reduce the level of intrinsic motivation or damage social norms of behavior. Our results show, however, that for tasks with relatively low levels of prior engagement the opposite can hold. While we could not directly compare the effects of the temporary scheme with a permanent bonus for sales leads, our back-of-the-envelope calculations suggest that the temporary incentive may be more profitable than a permanent scheme when leading to persistent changes of behavior without coming with additional costs in the long-term.

Our results have several implications for managerial practice. For one, we provide evidence that habit formation, which often has been discussed in the context of consumption habits or health-related behaviors, also matters in the workplace. Once employees have acquired a habit they may continue to follow it even without further external impulses. While standard incentive theory typically suggests that it is crucial to use stable bonus plans that pay employees based on value created, our results thus also imply that firms may well consider changing the incentivized objectives from time to time targeting very specific tasks, thereby guiding employees to adopt novel routines supporting strategic objectives.²⁷

²⁷See [Manthei, Sliwka and Vogelsang \(2021\)](#) for a related argument in the context of learning-by-doing.

Taken together, even short-term incentive programs can have a lasting impact by fostering productive habits that persist beyond the incentive period. Such initiatives can help organizations align employee behavior with strategic goals, equipping managers during periods of transition with an additional tool to navigate evolving circumstances.

References

- Acland, Dan, and Matthew R Levy.** 2015. "Naiveté, projection bias, and habit formation in gym attendance." *Management Science*, 61(1): 146–160.
- Alfitian, Jakob, Dirk Sliwka, and Timo Vogelsang.** 2024. "When bonuses backfire: Evidence from the workplace." *Management Science*, 70(9): 6395–6414.
- Allcott, Hunt, and Todd Rogers.** 2014. "The short-run and long-run effects of behavioral interventions: Experimental evidence from energy conservation." *American Economic Review*, 104(10): 3003–3037.
- Andon, Paul, Clinton Free, Radzi Jidin, Gary S Monroe, and Michael J Turner.** 2018. "The impact of financial incentives and perceptions of seriousness on whistleblowing intention." *Journal of Business Ethics*, 151: 165–178.
- Arrow, Kenneth J.** 1962. "The economic implications of learning by doing." *The Review of Economic Studies*, 29(3): 155–173.
- Autor, David, Caroline Chin, Anna Salomons, and Bryan Seegmiller.** 2024. "New frontiers: The origins and content of new work, 1940–2018." *The Quarterly Journal of Economics*, 139(3): 1399–1465.
- Autor, David H, Frank Levy, and Richard J Murnane.** 2003. "The skill content of recent technological change: An empirical exploration." *The Quarterly Journal of Economics*, 118(4): 1279–1333.
- Bandiera, Oriana, Iwan Barankay, and Imran Rasul.** 2011. "Field experiments with firms." *Journal of Economic Perspectives*, 25(3): 63–82.
- Becker, Gary S.** 1965. "A Theory of the Allocation of Time." *The Economic Journal*, 75(299): 493–517.
- Becker, Gary S, and Kevin M Murphy.** 1988. "A theory of rational addiction." *Journal of Political Economy*, 96(4): 675–700.
- Bénabou, Roland, and Jean Tirole.** 2003. "Intrinsic and extrinsic motivation." *The Review of Economic Studies*, 70(3): 489–520.
- Berger, Leslie, Stephen Perreault, and James Wainberg.** 2017. "Hijacking the moral imperative: How financial incentives can discourage whistleblower reporting." *Auditing: A Journal of Practice & Theory*, 36(3): 1–14.
- Berger, Philip G, Wei Cai, Lin Qiu, and Cindy Xinyi Shen.** 2024. "Employer and Employee Responses to Generative AI: Early Evidence." *Available at SSRN 4874061*.
- Böke, Jens, Daniela De la Parra, John Gallemore, and Stephen Glaeser.** 2024. "Artificial Intelligence and White-Collar Work: Evidence from the Accounting Profession." *Available at SSRN 5039640*.

- Brink, Alisa G, D Jordan Lowe, and Lisa M Victoravich.** 2013. "The effect of evidence strength and internal rewards on intentions to report fraud in the Dodd-Frank regulatory environment." *Auditing: A Journal of Practice & Theory*, 32(3): 87–104.
- Buell, Ryan W, Wei Cai, and Tatiana Sandino.** 2022. "Learning or playing? The effect of gamified training on performance." *Harvard Business School Technology & Operations Mgt. Unit Working Paper*.
- Buyalskaya, Anastasia, Hung Ho, Katherine L Milkman, Xiaomin Li, Angela L Duckworth, and Colin Camerer.** 2023. "What can machine learning teach us about habit formation? Evidence from exercise and hygiene." *Proceedings of the National Academy of Sciences*, 120(17): 1–7.
- Cardinaels, Eddy, and Huaxiang Yin.** 2015. "Think twice before going for incentives: Social norms and the principal's decision on compensation contracts." *Journal of Accounting Research*, 53(5): 985–1015.
- Casas-Arce, Pablo, and F Asis Martinez-Jerez.** 2009. "Relative performance compensation, contests, and dynamic incentives." *Management Science*, 55(8): 1306–1320.
- Celhay, Pablo A, Paul J Gertler, Paula Giovagnoli, and Christel Vermeersch.** 2019. "Long-run effects of temporary incentives on medical care productivity." *American Economic Journal: Applied Economics*, 11(3): 92–127.
- Chapman, Gretchen, and Uri Gneezy.** 2024. "Change behavior, motivation Will follow (?): Acquired taste and incentives." *AEA Papers and Proceedings*, 114: 660–665.
- Charness, Gary, and Uri Gneezy.** 2009. "Incentives to exercise." *Econometrica*, 77(3): 909–931.
- Choi, Jongwoon, Gary W Hecht, and William B Tayler.** 2012. "Lost in translation: The effects of incentive compensation on strategy surrogation." *The Accounting Review*, 87(4): 1135–1163.
- Choi, Jongwoon, Gary W Hecht, and William B Tayler.** 2013. "Strategy selection, surrogation, and strategic performance measurement systems." *Journal of Accounting Research*, 51(1): 105–133.
- Danilov, Anastasia, and Dirk Sliwka.** 2017. "Can contracts signal social norms? Experimental evidence." *Management Science*, 63(2): 459–476.
- De Charms, Richard.** 1968. *Personal Causation: The Internal Affective Determinants of Behavior*. Routledge.
- Deci, Edward L.** 1971. "Effects of externally mediated rewards on intrinsic motivation." *Journal of personality and Social Psychology*, 18(1): 105.
- Deci, Edward L, Richard Koestner, and Richard M Ryan.** 1999. "The undermining effect is a reality after all—Extrinsic rewards, task interest, and self-determination: Reply to Eisenberger, Pierce, and Cameron (1999) and Lepper, Henderlong, and Gingras (1999)." *Psychological Bulletin*, 125(6): 692–700.

- Dickinson, Anthony.** 1985. "Actions and habits: the development of behavioural autonomy." *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 308(1135): 67–78.
- Eisenhardt, Kathleen M.** 1989. "Agency theory: An assessment and review." *Academy of Management Review*, 14(1): 57–74.
- Eyring, Henry, and VG Narayanan.** 2018. "Performance effects of setting a high reference point for peer-performance comparison." *Journal of Accounting Research*, 56(2): 581–615.
- Festinger, Leon.** 1957. *A Theory of Cognitive Dissonance*. Stanford University Press.
- Festinger, Leon, and James M Carlsmith.** 1959. "Cognitive consequences of forced compliance." *The Journal of Abnormal and Social Psychology*, 58(2): 203.
- Frey, Bruno S, and Felix Oberholzer-Gee.** 1997. "The cost of price incentives: An empirical analysis of motivation crowding-out." *The American Economic Review*, 87(4): 746–755.
- Friebel, Guido, Matthias Heinz, Miriam Krueger, and Nikolay Zubanov.** 2017. "Team incentives and performance: Evidence from a retail chain." *American Economic Review*, 107(8): 2168–2203.
- Gallani, Susanna.** 2023. "Conduit incentives: Eliciting cooperation from workers outside of managers' control." *The Accounting Review*, 98(3): 175–202.
- Gardner, Benjamin, Amanda L Rebar, and Phillippa Lally.** 2022. "How does habit form? Guidelines for tracking real-world habit formation." *Cogent Psychology*, 9(1): 2041277.
- Gardner, Benjamin, Charles Abraham, Phillippa Lally, and Gert-Jan de Bruijn.** 2012. "Towards parsimony in habit measurement: Testing the convergent and predictive validity of an automaticity subscale of the Self-Report Habit Index." *International Journal of Behavioral Nutrition and Physical Activity*, 9: 1–12.
- Gardner, Benjamin, Gert-Jan de Bruijn, and Phillippa Lally.** 2011. "A systematic review and meta-analysis of applications of the self-report habit index to nutrition and physical activity behaviours." *Annals of Behavioral Medicine*, 42(2): 174–187.
- Gertler, Paul, Sean Higgins, Aisling Scott, and Enrique Seira.** 2018. "The long-term effects of temporary incentives to save: Evidence from a prize-linked savings field experiment." Abdul Latif Jameel Poverty Action Lab.
- Gibbs, Michael, and Sergei Bazylik.** 2022. "How is new technology changing job design?" *IZA World of Labor* No 344v2.
- Gillan, Claire M, A Ross Otto, Elizabeth A Phelps, and Nathaniel D Daw.** 2015. "Model-based learning protects against forming habits." *Cognitive, Affective, & Behavioral Neuroscience*, 15(3): 523–536.
- Gneezy, Uri, and Aldo Rustichini.** 2000a. "A fine is a price." *The Journal of Legal Studies*, 29(1): 1–17.

- Gneezy, Uri, and Aldo Rustichini.** 2000b. "Pay enough or don't pay at all." *The Quarterly Journal of Economics*, 115(3): 791–810.
- Gneezy, Uri, Stephan Meier, and Pedro Rey-Biel.** 2011. "When and why incentives (don't) work to modify behavior." *Journal of Economic Perspectives*, 25(4): 191–210.
- Grabe, Leonhard, and Dirk Sliwka.** 2025. "Managing Skills in Organizations - Evidence from a Field Experiment." *IZA Discussion Paper No 17727*.
- Grabner, Isabella, and Frank Moers.** 2013. "Managers' choices of performance measures in promotion decisions: An analysis of alternative job assignments." *Journal of Accounting Research*, 51(5): 1187–1220.
- Harmon-Jones, Eddie, and Judson Mills.** 2019. *An introduction to cognitive dissonance theory and an overview of current perspectives on the theory*. American Psychological Association.
- Holmström, Bengt.** 1979. "Moral hazard and observability." *The Bell Journal of Economics*, 10(1): 74–91.
- Holmström, Bengt.** 1999. "Managerial incentive problems: A dynamic perspective." *The Review of Economic Studies*, 66(1): 169–182.
- Holmström, Bengt.** 2017. "Pay for performance and beyond." *American Economic Review*, 107(7): 1753–1777.
- Huffman, David, and Michael Bognanno.** 2018. "High-powered performance pay and crowding out of nonmonetary motives." *Management Science*, 64(10): 4669–4680.
- Hyun, Jeong-Hoon, Michal Matějka, Peter Oh, and Tae Sik Ahn.** 2022. "Performance targets and ex post incentive plan adjustments." *Contemporary Accounting Research*, 39(2): 863–892.
- Jovanovic, Boyan, and Yaw Nyarko.** 1996. "Learning by Doing and the Choice of Technology." *Econometrica*, 64(6): 1299–1310.
- Kunz, Alexis H, and Dieter Pfaff.** 2002. "Agency theory, performance evaluation, and the hypothetical construct of intrinsic motivation." *Accounting, Organizations and Society*, 27(3): 275–295.
- Larcom, Shaun, Ferdinand Rauch, and Tim Willems.** 2017. "The benefits of forced experimentation: Striking evidence from the London underground network." *The Quarterly Journal of Economics*, 132(4): 2019–2055.
- Lazear, Edward P.** 2000. "Performance pay and productivity." *American Economic Review*, 90(5): 1346–1361.
- Lazear, Edward P.** 2018. "Compensation and incentives in the workplace." *Journal of Economic Perspectives*, 32(3): 195–214.
- Lepper, Mark R, David Greene, and Richard E Nisbett.** 1973. "Undermining children's intrinsic interest with extrinsic reward: A test of the "overjustification" hypothesis." *Journal of Personality and Social Psychology*, 28(1): 129.

- Loewenstein, George, and Erik Angner.** 2003. "Predicting and indulging changing preferences." In *Time and Decision: Economic and Psychological Perspectives on Intertemporal Choice.* , ed. George Loewenstein, Daniel Read and Roy F. Baumeister, Chapter 12, 351–391. Russell Sage Foundation New York.
- Mahlendorf, Matthias D., and Timo Vogelsang.** 2024. "Performance measures and incentives: evidence from field experiments in firms." In *Research Handbook on Performance Measurement for Management Control. Chapters,* , ed. Anne M. Lillis and Jennifer Grafton, Chapter 8, 155–166. Edward Elgar Publishing.
- Manthei, Kathrin, Dirk Sliwka, and Timo Vogelsang.** 2021. "Performance pay and prior learning—evidence from a retail chain." *Management Science*, 67(11): 6998–7022.
- Manthei, Kathrin, Dirk Sliwka, and Timo Vogelsang.** 2023. "Information, incentives, and attention: A field experiment on the interaction of management controls." *The Accounting Review*, 98(5): 455–479.
- Mazar, Asaf, and Wendy Wood.** 2018. "Defining habit in psychology." In *The Psychology of Habit: Theory, Mechanisms, Change, and Contexts.* , ed. Bas Verplanken, 13–29. Springer.
- Meier, Stephan.** 2007. "Do Subsidies Increase Charitable Giving in the Long Run? Matching Donations in a Field Experiment." *Journal of the European Economic Association*, 5(6): 1203–1222.
- Melchionne, Kevin.** 2007. "Acquired taste." *Contemporary Aesthetics (Journal Archive)*, 5(Article 11).
- Merchant, Kenneth, and Wim Van der Stede.** 2017. *Management Control Systems: Performance Measurement, Evaluation and Incentives.* Pearson Deutschland.
- Mihaylov, Emil, and Kea Gartje Tijdens.** 2019. "Measuring the routine and non-routine task content of 427 four-digit ISCO-08 occupations." Tinbergen Institute discussion paper.
- Prendergast, Canice.** 1999. "The provision of incentives in firms." *Journal of economic literature*, 37(1): 7–63.
- Ross, Stephen A.** 1973. "The economic theory of agency: The principal's problem." *The American Economic Review*, 63(2): 134–139.
- Royer, Heather, Mark Stehr, and Justin Sydnor.** 2015. "Incentives, commitments, and habit formation in exercise: evidence from a field experiment with workers at a fortune-500 company." *American Economic Journal: Applied Economics*, 7(3): 51–84.
- Ryan, Richard M., Valerie Mims, and Richard Koestner.** 1983. "Relation of reward contingency and interpersonal context to intrinsic motivation: A review and test using cognitive evaluation theory." *Journal of Personality and Social Psychology*, 45(4): 736–750.
- Sliwka, Dirk.** 2007. "Trust as a signal of a social norm and the hidden costs of incentive schemes." *American Economic Review*, 97(3): 999–1012.

- Stigler, George J, and Gary S Becker.** 1977. "De gustibus non est disputandum." *American Economic Review*, 67(2): 76–90.
- Verplanken, Bas, and Sheina Orbell.** 2003. "Reflections on past behavior: a self-report index of habit strength 1." *Journal of Applied Social Psychology*, 33(6): 1313–1330.
- Verplanken, Bas, and Sheina Orbell.** 2022. "Attitudes, habits, and behavior change." *Annual Review of Psychology*, 73(1): 327–352.
- Volpp, Kevin G, and George Loewenstein.** 2020. "What is a habit? Diverse mechanisms that can produce sustained behavior change." *Organizational Behavior and Human Decision Processes*, 161: 36–38.
- Wang, Kelly K, Mandy M Cheng, and Linda J Chang.** 2023. "Reducing strategy surrogation: The effects of performance measurement system flexibility and environmental dynamism." *The Accounting Review*, 98(4): 435–456.
- Wood, Wendy, and David T Neal.** 2009. "The habitual consumer." *Journal of Consumer Psychology*, 19(4): 579–592.
- Wood, Wendy, and Dennis R  nger.** 2016. "Psychology of habit." *Annual Review of Psychology*, 67: 289–314.
- Wood, Wendy, Jeffrey M Quinn, and Deborah A Kashy.** 2002. "Habits in everyday life: thought, emotion, and action." *Journal of Personality and Social Psychology*, 83(6): 1281.
- Wyatt, Anne, and Hermann Frick.** 2010. "Accounting for investments in human capital: A review." *Australian Accounting Review*, 20(3): 199–220.
- Yang, Nan, and Yong Long Lim.** 2018. "Temporary incentives change daily routines: Evidence from a field experiment on Singapore's subways." *Management Science*, 64(7): 3365–3379.

A Appendix

Table 6: Check for Selective Attrition

	<i>Incentive Group</i>		<i>Control Group</i>		p-value
	Mean	S.D.	Mean	S.D.	
Attrition	0.098	0.297	0.106	0.308	0.671
Attrition Incentive Phase	0.046	0.209	0.048	0.213	0.882
Attrition 4 Weeks After	0.013	0.113	0.006	0.080	0.313
Weekly Leads Pre (Left)	0.886	1.504	0.907	1.646	0.950
Bonuses Pre (Left)	0.311	0.633	0.367	0.698	0.683
Observations	461		462		923

Note: This table reports summary statistics for the individuals who left during our observation period separately for the incentive and the control group. We also report p -values for tests on the equality of proportions for the attrition shares as well as for t-tests for the continuous variables. Weekly leads in the pre-period as well as the number of bonuses individuals could have earned in the pre-period are only compared for the subsample of individuals that left during our observation period.

Table 7: Effect on Number of (Successful) Sales Leads (Cross-Sectional)

	Incentive Period		Post-Incentive Period	
	<i>Sales Leads_i</i> (1)	<i>Successful Leads_i</i> (2)	<i>Sales Leads_i</i> (3)	<i>Successful Leads_i</i> (4)
<i>Treat_i</i>	0.393*** (0.073)	0.086*** (0.033)	0.201** (0.102)	0.071** (0.036)
Company Fixed Effects	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	829	829	829	829
Adjusted R-squared	0.749	0.619	0.624	0.600

Note: This table reports results of a regression of the number of reported sales leads (columns (1) and (3)) or successful sales leads (columns (2) and (4)) on being part of the treatment group, i.e. incentive group, during (columns (1) and (2)) as well as after (columns (3) and (4)) the incentive period. The data are at the individual level. For each individual, sales leads and successful sales leads are measured as the mean over the weeks within the respective period. The regression is estimated on the balanced panel, i.e., all individuals who remain in the sample at the end of the observation period. We include company fixed effects and control for pre-treatment means of individual weekly sales leads and successful sales leads. Robust standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 8: Effect on Number of (Successful) Sales Leads (All Assigned Individuals)

	<i>Sales Leads_{it}</i> (1)	<i>Successful Leads_{it}</i> (2)
<i>Treat_i × Incentive_t</i>	0.368*** (0.070)	0.085*** (0.033)
<i>Treat_i × Post Incentive_t</i>	0.150 (0.095)	0.062* (0.035)
<i>p</i> -value Inc=Post	0.004	0.403
Time Fixed Effects	Yes	Yes
Individual Fixed Effects	Yes	Yes
Clustered at	Individual	Individual
Number of Clusters	923	923
Observations	30,459	30,459
Adjusted R-squared	0.565	0.427

Note: This table reports results of a difference-in-differences regression of the number of reported sales leads (column (1)) or successful sales leads (column (2)) on being part of the treatment group, i.e. incentive group, during as well as after the incentive phase in comparison to the pre-incentive phase. The data is on the individual-week level. The regression is based on all individuals who were randomly assigned before the start of the incentive phase. We include individual as well as week fixed effect. Standard errors are clustered at the individual level, and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 9: Effect on Number of Weeks with Leads

	<i>Total Bonuses_{it}</i> (1)	<i>Weeks with Leads_{it}</i> (2)
$Treat_i \times Incentive_t$	0.225*** (0.060)	0.880*** (0.185)
$Treat_i \times Post\ Incentive_t$	0.044 (0.058)	0.487*** (0.187)
<i>p</i> -value Inc=Post	0.003	0.022
Phase Fixed Effects	Yes	Yes
Individual Fixed Effects	Yes	Yes
Clustered at	Individual	Individual
Number of Clusters	829	829
Observations	2,487	2,487
Adjusted R-squared	0.529	0.766

Note: This table reports results of a difference-in-differences regression of the number of bonuses an individual would have gotten (column 1) or number of weeks with at least one sales lead (column 2) on being part of the treatment group, i.e. incentive group, during as well as after the incentive phase in comparison to the pre-incentive phase. The data is on the individual-phase level. The regression is based on the balanced panel, i.e. all individuals who are still there at the end of the observation period. We include individual and phase fixed effects. Standard errors are clustered at the individual level, and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 10: Survey Results (Subset of Participants Answering All Items)

	<i>Automaticity</i>		<i>Intrinsic</i>			<i>Importance</i>		<i>Communication</i>	
	<i>Leads_i</i> (1)	<i>Calls_i</i> (2)	<i>Leads_i</i> (3)	<i>Calls_i</i> (4)	<i>Knowledge_i</i> (5)	<i>Leads_i</i> (6)	<i>Calls_i</i> (7)	<i>All_i</i> (8)	<i>Leads_i</i> (9)
<i>Treat_i</i>	0.125 (0.168)	0.234 (0.163)	0.338** (0.155)	0.458*** (0.150)	0.050 (0.184)	0.039 (0.179)	0.059 (0.168)	0.149 (0.180)	0.124 (0.176)
Observations	128	128	128	128	128	128	128	128	128
R-squared	0.004	0.015	0.037	0.069	0.001	0.000	0.001	0.006	0.004

Note: This table presents the results of a regression analysis examining the effect of being in the treatment group (i.e., the incentive group) on various survey constructs. The constructs measured in the survey include automaticity, intrinsic motivation, knowledge, perceived task importance, and communication frequency. Each construct is measured by calculating the mean of multiple survey items and then standardized using z-scores. Columns (1), (3), and (6) refer to sales leads. Columns (2), (4), and (7) correspond to another customer-oriented measure, specifically the requested calls made before customer visits. Column (5) reports the results related to knowledge on how to generate sales leads. Column (8) refers to the frequency of communication with the supervisor, while Column (9) focuses on how often that communication was about sales. The data is at the individual level and includes only individuals who answered all survey items. Robust standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 11: Survey Results (With Other Task)

	<i>Automaticity</i>		<i>Intrinsic</i>			<i>Importance</i>		<i>Communication</i>	
	<i>Leads_i</i> (1)	<i>Calls_i</i> (2)	<i>Leads_i</i> (3)	<i>Calls_i</i> (4)	<i>Knowledge_i</i> (5)	<i>Leads_i</i> (6)	<i>Calls_i</i> (7)	<i>All_i</i> (8)	<i>Leads_i</i> (9)
<i>Treat_i</i>	0.115 (0.127)	0.105 (0.125)	0.315** (0.126)	0.328** (0.127)	0.005 (0.126)	0.166 (0.126)	0.151 (0.126)	0.079 (0.126)	0.105 (0.126)
Observations	248	253	245	243	253	252	253	253	253
R-squared	0.003	0.003	0.025	0.027	0.000	0.007	0.006	0.002	0.003

Note: This table presents the results of a regression analysis examining the effect of being in the treatment group (i.e., the incentive group) on various survey constructs. The constructs measured in the survey include automaticity, intrinsic motivation, knowledge, perceived task importance, and communication frequency. Each construct is measured by calculating the mean of multiple survey items and then standardized using z-scores. Columns (1), (3), and (6) refer to sales leads. Columns (2), (4), and (7) correspond to another customer-oriented measure, specifically the requested calls made before customer visits. Column (5) reports the results related to knowledge on how to generate sales leads. Column (8) refers to the frequency of communication with the supervisor, while Column (9) focuses on how often that communication was about sales. The data is at the individual level. There are slight differences in the number of observations across the constructs, as not all survey participants answered every item. Robust standard errors are reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 12: Effect on Sales Profit

	<i>Individual Sales Profit_{it}</i>	
	<i>Gross</i> (1)	<i>Net (With Bonus)</i> (2)
$Treat_i \times Incentive_t$	14.527 (17.931)	-13.999 (17.776)
$Treat_i \times Post\ Incentive_t$	23.421 (20.420)	23.421 (20.420)
<i>p</i> -value Inc=Post	0.585	0.022
Time Fixed Effects	Yes	Yes
Fixed Effects	Individual	Individual
Clustered at	Individual	Individual
Number of Clusters	829	829
Observations	6,632	6,632
Adjusted R-squared	0.582	0.579

Note: This table reports results of a difference-in-differences regression of the individual employee gross sales profit (column (1)) or the net sales profit, i.e. subtracting bonus payments (column (2)) on being part of the treatment group, i.e. incentive group, during as well as after the incentive phase in comparison to the pre-incentive phase. The data is at the individual-month level. The regression is based on the balanced panel, i.e. only includes individuals who are still there at the end of our observation period. We include individual as well as month fixed effect. Standard errors are clustered at the individual level, and reported in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Dear colleagues,

We want to inspire our customers with the best service again this year. To make it worthwhile for you to stay on track with your sales activities, *MultiCo* will soon start an initiative that targets lead generation. As not everyone can take part in the initiative at the same time a draw was held in advance to determine who can participate when.

Congratulations, you are in the first group!

We will launch a 12-week initiative for you starting Monday, 06.03.2023: Connect Four!

Those who regularly engage with customers and submit lead can receive up to €300 in additional bonuses.

How does “Connect Four” work?

If you manage to submit at least one sales lead per week for four consecutive weeks, you will receive an additional bonus of €100¹. If the streak is interrupted, it must be started again from the beginning. Those who manage to complete multiple four-week streaks can look forward to an even higher bonus.

What is incentivized?

Every lead that is submitted with the customer’s consent and recorded on the *platform name* counts. This includes, for example, *product name*, *product name*, *product name*, or *product name*. Unlike previous initiatives, leads are not tied to a successful sale. However, all current bonuses still apply – so it’s doubly worthwhile for you to submit a promising lead. You can find more details about the bonuses in the attached description.

Would you like to brush up on your knowledge on how to generate sales leads?

Then take a look at our online course on lead generation for field service technicians! Here you will not only learn how lead generation works on a technical level, but also how you can generate leads more easily and without abandoning your identity as a technician.

The course takes place every Tuesday from 17:00 - 18:30 and you can register for a day of your choice via this link

[link]

Figure 4: Wording of the Email Announcement (Treatment Group)

Note: This figure shows the English translation of the email announcement for the employees in the treatment group. For confidentiality reasons, the real name of our partner company has been replaced with ‘MultiCo’. Product and platform names have also been replaced.

CONNECT FOUR: EXAMPLES

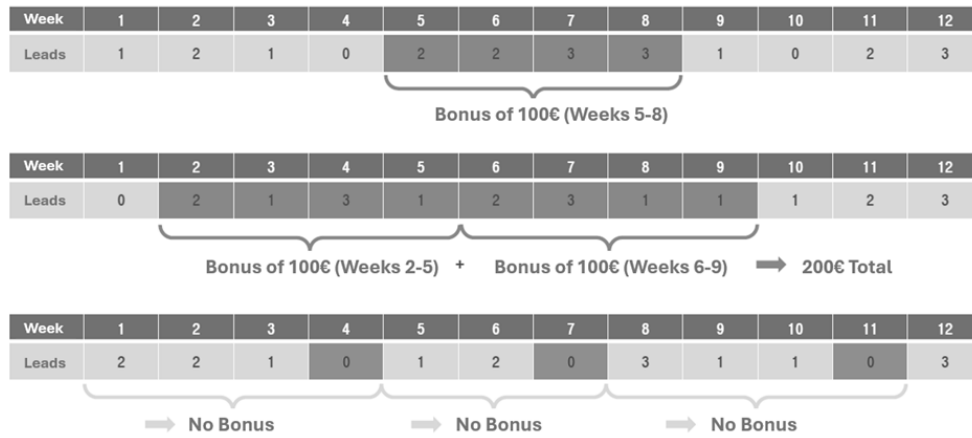


Figure 5: Email Appendix (Treatment Group)

Note: This figure shows the English translation of the appendix to the email announcement for the employees in the treatment group.

Dear colleagues,

We want to inspire our customers with the best service again this year. To make it worthwhile for you to stay on track with your sales activities, *MultiCo* will soon start an initiative that targets lead generation. As not everyone can take part in the initiative at the same time, a draw was held in advance to determine who can participate when.

As soon as the initiative becomes relevant for you, you will be informed by us in good time!

Would you like to brush up on your knowledge on how to generate sales leads?

Then take a look at our online course on lead generation for field service technicians! Here you will not only learn how lead generation works on a technical level, but also how you can generate leads more easily and without abandoning your identity as a technician.

The course takes place every Tuesday from 17:00 - 18:30 and you can register for a day of your choice via this link:

[Link]

Figure 6: Wording of the Email Announcement (Control Group)

Note: This figure shows the English translation of the email announcement for the employees in the control group. For confidentiality reasons, the real name of our partner company has been replaced with 'MultiCo'.